



S T I X O R

Water Breakthrough POC

Minutes of Meeting

Prepared for:

MARI Energies

Date:

April, 8 2026

Prepared by:

Stixor Technologies



Minutes of Meeting

Meeting	Stixor <> Mari Energies — Water Breakthrough POC: Data Review & Kickoff
Date	April 8, 2026
Time	11:00 AM PKT

1. Meeting objectives

- Review the production and subsurface data shared by Mari Energies on April 6 and April 7
- Walk through the data gap analysis prepared by Stixor
- Agree on the items still required, and align on the POC scope, scale, and timeline

2. Discussion summary

2.1. Data Review

Stixor walked Mari through the gap analysis checklist covering all production, pressure, rock, completion, geometry, geology and fluid fields against the original Section 6 data requirements. Mari acknowledged the gaps identified and confirmed they will provide the missing items listed under Action Items below.

2.2. Choke Setting Records (Pre-Aug 2015)

Stixor flagged that choke records are only available from Aug 2015 onwards, leaving wells without choke history at the time of breakthrough. Mari to share pre-2015 records. Where records cannot be retrieved, Stixor will impute using forward-filling and average values. The list of those 17 wells is given below:

M-11-HRL, M-13-HRL, M-22-HRL, M-41-HRL, M-50-HRL, M-51-HRL, M-56-HRL, M-57-HRL, M-58-HRL, M-61-HRL, M-63-HRL, M-65-HRL, M-67-HRL, M-75-HRL, M-81-HRL, M-82-HRL, M-E-2-HRL

2.3. Pre-1990 Wellhead Flowing Pressure

WHFP for following wells only began in June 1990, leaving the first ~12 years of production without pressure data.

M-11-HRL, M-13-HRL, M-22-HRL, M-41-HRL, M-50-HRL, M-51-HRL

2.4. Line Pressure Gap

Line Pressure is currently ~38% complete. Mari to share the remaining records. Where Line Pressure remains missing, Stixor will compute drawdown (WHFP minus Line Pressure) only for months where both values exist, and use WHFP alone otherwise.

2.5. Bottomhole and Initial Reservoir Pressure

Direct BHP measurements are not available. Mari agreed to share estimates for both BHP per well and Initial Reservoir Pressure (P_i) at the start of production.

2.6. Well Coordinates and Trajectories

This is the highest-priority unblocking item. Mari to share:

- Surface coordinates (X/Y or lat-long) for all 22 wells
- Bottomhole coordinates for all wells, including vertical wells
- Full trajectories

Coordinates and trajectories enable neighbour-based features, distance-to-OWC calculations, and analog matching between the new horizontals and the legacy verticals.

2.7. Gas-Water Contact (GWC) Depth

Mari to share the GWC depth for the Habib Rahi Limestone reservoir. A single field-wide value is acceptable as a starting point. This is the single most important input for the physics-informed coning features; vertical distance from perforations (in TVD) to GWC drives the cone-rise hazard term in the model.

2.8. Gas Z-factor and Gas Gravity

Mari to share Z-factor and gas gravity (specific gravity) for the reservoir. These enable conversion of WHFP to bottomhole pressure via standard correlations (e.g., Cullender-Smith), and allow construction of p/z vs cumulative gas plots for material-balance feature engineering.

2.9. Event label definition

The water breakthrough event label will initially be defined using Water-Gas Ratio ($WGR > 5$ bbl/MMcf, sustained). In a later iteration, the event definition will be refined using chloride concentration to distinguish true formation-water breakthrough from condensed water or completion flowback.

2.10. POC Scope

Reservoir: Habib Rahi Limestone only (single reservoir, not multi-zone)

- Dry-test wells: 5 wells designated by Mari for forecast validation
- Modelling approach: Physics-informed survival analysis (Weibull AFT as baseline, with Cox PH and Random Survival Forest for benchmarking)
- Output: P10 / P50 / P90 water breakthrough timing per dry-test well
- Validation: Physics-compliance checks (higher rate, higher Sw , closer to GWC must shorten time-to-breakthrough)

2.11. Timeline

- Kickoff: April 9, 2026 (*subject to provision of missing data*)
- Duration: 2 weeks
- Status updates: Weekly (*to be scheduled in the week by Stixor*)
- Final delivery: April 23, 2026

3. Decisions

1. Mari will provide all 7 outstanding data items by April 9, 2026
2. POC will run for 2 weeks starting April 9, 2026, given the missing items are provided with weekly status updates.
3. POC scope is restricted to a single reservoir (Habib Rahi Limestone).
4. Dry-testing will be performed on 5 wells designated by Mari.
5. Where data items cannot be retrieved, Stixor will apply the imputation / fallback strategies discussed during the meeting and document them in the final report.

4. Action items

#	Action
1	Share pre-Aug 2015 choke setting records for the 17 legacy wells
2	Share pre-June 1990 Wellhead Flowing Pressure (WHFP) records for M-11 and M-13
3	Share remaining Line Pressure records to fill the gap currently in the dataset
4	Provide engineering estimates for Bottomhole Pressure (BHP) and Initial Reservoir Pressure (P _i) per well
5	Provide well coordinates: surface + bottomhole for vertical wells; surface + bottomhole + full trajectories for the 5 horizontal wells
6	Provide Gas-Water Contact (GWC) depth (single value per reservoir is acceptable)
7	Share Gas Z-factor and Gas gravity (PVT) for the reservoir